

Solver Specification:

highOrderElectroActivationFoamImplicit

A high-order, implicit Picard / diagonal-Ion linearised monodomain solver in OpenFOAM
with an Eigen linear-algebra back-end

Pablo

May 13, 2026

Abstract

This report specifies the algorithms implemented in the OpenFOAM solver **highOrderElectroActivationFoamImplicit**. The solver advances the monodomain equations of cardiac electrophysiology coupled to the ten Tusscher–Noble–Noble–Panfilov (TNNP, 2004) ionic cell model. The diffusion operator is discretised with the cell-centred finite-volume framework of OpenFOAM, optionally upgraded to a Local Radial-Extrapolation (LRE) high-order reconstruction; the ionic source is evaluated either at cell centres or at LRE cell quadrature points; time integration follows a θ -scheme (backward Euler or Crank–Nicolson); the resulting *nonlinear* coupled system is solved with a fixed-point Picard iteration that optionally augments the diagonal of the implicit operator with a finite-difference approximation of $\partial I_{\text{ion}} / \partial V_m$ (the *diagonal-Ion* variant). The sparse linear system is factored or iteratively solved with Eigen (**SparseLU** or **BiCGSTAB+ILUT**); the per-cell ionic ODE uses OpenFOAM’s adaptive RKF45 integrator with configurable tolerances; the TNNP cell model is wrapped in a phase-aware **protectState** guard that enforces physical bounds before every algebraic evaluation; activation times are recorded by linear interpolation across the first threshold crossing and benchmark points are sampled by inverse-distance weighting. All algorithmic parameters are exposed through the **constant/spatialIntegrationProperties** and **constant/timeIntegrationProperties** dictionaries, and the companion Python driver **run.convergence.py** generates these dictionaries from a single configuration file. Section 12 collects measured P8 activation times, wall-clock cost and peak memory from four independent result trees of the Niederer 2012 benchmark.

Contents

1	Notation	2
2	Governing equations	3
2.1	Monodomain PDE	3
2.2	Ionic ODE	3
2.3	External stimulus	3
3	Time discretisation: the θ-scheme	3
4	Spatial discretisation	4
4.1	Standard orthogonal FV Laplacian	4
4.2	High-order LRE diffusion	4
4.3	Mass matrices	4
4.4	High-order reconstruction at ionic quadrature points	4
5	Nonlinear solver	4
5.1	Picard (default)	4
5.2	Diagonal-Ion linearised Picard	5

5.3	Convergence criteria	5
5.4	Rollback on non-convergence	5
6	Linear solver: Eigen back-end	5
7	Ionic ODE integration	5
8	TNNP numerical protection	6
9	Activation tracking and benchmarks	6
10	Solver flow chart	6
11	Driver script: run_convergence.py	7
11.1	Geometry and parameter sweep	7
11.2	Nonlinear and linear solver	8
11.3	Inner ODE integrator (RK45)	8
11.4	Post-processing and run-control	8
12	Niederer benchmark results	9
12.1	Results for $\Delta t = 1.0 \cdot 10^{-3}$ s	9
12.2	Results for $\Delta t = 1.0 \cdot 10^{-4}$ s	10
12.3	Results for $\Delta t = 1.0 \cdot 10^{-5}$ s	11
12.4	Results for $\Delta t = 1.0 \cdot 10^{-6}$ s	13
A	Dimension table	14

1 Notation

Symbol	Meaning	Units
$\Omega \subset \mathbb{R}^d$	Spatial domain ($d \in \{2, 3\}$)	m
$\partial\Omega$	Insulated boundary (zero-flux Neumann)	—
$t \in (0, T]$	Time	s
$V_m(\mathbf{x}, t)$	Transmembrane voltage	V
$\mathbf{s}(\mathbf{x}, t) \in \mathbb{R}^{17}$	TNNP ionic states	mixed
$I_{\text{ion}}(V_m, \mathbf{s})$	Total ionic current density	V s^{-1}
$I_{\text{ext}}(\mathbf{x}, t)$	External stimulus current density	A m^{-3}
χ	Membrane surface-to-volume ratio	m^{-1}
C_m	Membrane capacitance per unit area	F m^{-2}
$\mathbf{D}(\mathbf{x})$	Bulk conductivity tensor	S m^{-1}
$\theta \in \{0.5, 1\}$	Time-stepping parameter (CN or BE)	—
Δt	Outer time-step length	s
N_c	Number of mesh cells	—
$\mathbf{V}^n \in \mathbb{R}^{N_c}$	Vector of cell-averaged V_m at time t^n	V
$\mathbf{M}, \mathbf{K}, \mathbf{L}$	Mass, stiffness, scaled diffusion matrix	—
$\mathbf{A}, \mathbf{B}$	Implicit LHS / RHS operators of the θ -scheme	—
ω	Picard / Newton damping factor	—

2 Governing equations

2.1 Monodomain PDE

The bulk voltage is propagated by the standard monodomain reaction–diffusion equation [1, 2]:

$$\chi C_m \frac{\partial V_m}{\partial t} = \nabla \cdot (\mathbf{D} \nabla V_m) - I_{\text{ion}}(V_m, \mathbf{s}) + I_{\text{ext}} \quad \text{in } \Omega \times (0, T], \quad (1)$$

subject to insulated boundaries $\mathbf{n} \cdot (\mathbf{D} \nabla V_m) = 0$ on $\partial\Omega \times (0, T]$, and initial condition $V_m(\mathbf{x}, 0) = -84 \text{ mV}$. The conductivity tensor $\mathbf{D}$ is symmetric positive-definite and either uniform or read as a `volTensorField` from disk.

Dividing (1) by χC_m gives the form actually used inside the solver:

$$\frac{\partial V_m}{\partial t} = \mathcal{L} V_m + s(V_m, \mathbf{s}), \quad \mathcal{L} = \frac{1}{\chi C_m} \nabla \cdot (\mathbf{D} \nabla \cdot), \quad s = -\frac{I_{\text{ion}}}{\chi C_m} + \frac{I_{\text{ext}}}{\chi C_m}. \quad (2)$$

2.2 Ionic ODE

At every spatial sampling point the TNNP state vector $\mathbf{s} \in \mathbb{R}^{17}$ evolves according to

$$\frac{d\mathbf{s}}{dt} = \mathbf{f}_{\text{TNNP}}(V_m, \mathbf{s}; \text{tissue}), \quad \mathbf{s}(0) = \mathbf{s}_0, \quad (3)$$

where `tissue` selects `epicardialCells`, `mCells` or `endocardialCells`. The 17 tracked states are

$$\mathbf{s} = (V, K_i, Na_i, Ca_i, Xr_1, Xr_2, Xs, m, h, j, d, f, fCa, s, r, g, Ca_{SR}).$$

The total ionic current returned to the PDE is `Iion_cm` of TNNP [1].

2.3 External stimulus

$$I_{\text{ext}}(\mathbf{x}, t) = \begin{cases} I_0 & \mathbf{x} \in \Omega_{\text{stim}}^{(b)}, \quad t \in [t_0^{(b)}, t_0^{(b)} + \tau], \\ 0 & \text{otherwise,} \end{cases} \quad (4)$$

with Niederer-benchmark defaults $I_0 = 50\,000 \text{ A m}^{-3}$ and $\tau = 2 \text{ ms}$.

3 Time discretisation: the θ -scheme

Time is advanced from t^n to $t^{n+1} = t^n + \Delta t$ by

$$\frac{V^{n+1} - V^n}{\Delta t} = \theta \mathbf{L} V^{n+1} + (1 - \theta) \mathbf{L} V^n + \theta \mathbf{s}^{n+1} + (1 - \theta) \mathbf{s}^n. \quad (5)$$

<code>implicitScheme</code>	θ	Scheme
<code>backwardEuler</code>	1	First-order, L-stable
<code>crankNicolson</code>	1/2	Second-order, A-stable

After dividing by Δt the implementation forms

$$[\mathbf{M}/\Delta t - \theta \mathbf{L}] V^{n+1} = [\mathbf{M}/\Delta t + (1 - \theta) \mathbf{L}] V^n + \theta \mathbf{s}^{n+1} + (1 - \theta) \mathbf{s}^n, \quad (6)$$

with the two Eigen `SparseMatrix<scalar, RowMajor>` operators

$$\mathbf{A} = \mathbf{M}/\Delta t - \theta \mathbf{L}, \quad \mathbf{B} = \mathbf{M}/\Delta t + (1 - \theta) \mathbf{L}, \quad \mathbf{L} = \mathbf{K}/(\chi C_m). \quad (7)$$

$\mathbf{A}$ and $\mathbf{B}$ are rebuilt at every time step. When $\theta = 1$ (BE) the explicit half of the diffusion vanishes and $\mathbf{B} = \mathbf{M}/\Delta t$.

4 Spatial discretisation

4.1 Standard orthogonal FV Laplacian

`assembleStandardOrthogonalStiffnessMatrix` produces the classical 5/7-point Laplacian. Per internal face f ,

$$a_f = \frac{|\mathbf{S}_f| (\mathbf{n}_f \cdot \mathbf{D}_f \cdot \hat{\mathbf{d}}_f)}{|\mathbf{d}_f|}, \quad \mathbf{D}_f = \frac{1}{2} (\mathbf{D}_{\text{own}} + \mathbf{D}_{\text{nei}}), \quad (8)$$

yielding the four standard contributions to $\mathbf{K}$. `zeroGradient` and `empty` patches contribute nothing; `fixedValue` patches contribute only to the owner diagonal.

4.2 High-order LRE diffusion

When `useHighOrder_Vm = true` and $d > 1$, `assembleHighOrderStiffnessMatrix` integrates the flux on a face quadrature with reconstructed gradients:

$$\Phi_f = \sum_{q \in \mathcal{Q}_f} w_q |\mathbf{S}_f| \mathbf{n}_f \cdot \mathbf{D}_f \cdot \left(\sum_{j \in \mathcal{S}_f} \mathbf{g}_{f,q,j} V_j \right), \quad (9)$$

with $\mathcal{S}_f$ the LRE face stencil and $\mathbf{g}_{f,q,j} = \text{QRGradFaceGPCoeffs}$.

4.3 Mass matrices

Two options selectable via `massMatrix`:

Lumped. `assembleDiagonalMassMatrix` writes $\mathbf{M} = \mathbf{I}$.

Consistent high-order. `assembleConsistentMassMatrixH0` fills

$$\mathbf{M}[i, j] = \sum_{q \in \mathcal{Q}_i} \frac{w_q}{\sum_{q'} w_{q'}} \phi_j(\mathbf{x}_q), \quad (10)$$

where ϕ_j is the LRE Taylor-basis function of stencil cell j (value δ_{ij} + grad + Hessian + 3rd-order derivative).

4.4 High-order reconstruction at ionic quadrature points

If `useHighOrder_Iion = true` the ionic ODE is integrated at cell quadrature points; V_m is reconstructed there by the same Taylor expansion (`reconstructVmAtIionIntegrationPoints`). The ionic contribution to the cell average is then

$$I_{\text{ion},i} = \frac{\sum_{q \in \mathcal{Q}_i} w_q I_{\text{ion},q}}{\sum_{q \in \mathcal{Q}_i} w_q} \quad (\text{averageIionIntegrationPointsToCells}). \quad (11)$$

5 Nonlinear solver

The fully discrete equation is nonlinear in $\mathbf{V}^{n+1}$ through $I_{\text{ion}}(\mathbf{V}^{n+1}, \mathbf{s}^{n+1})$. Three closing strategies are implemented; `nonlinearMethod` selects between them.

5.1 Picard (default)

Given an iterate $\mathbf{x}_k$, the next iterate is the unique solution of the linear system

$$\mathbf{A} \mathbf{x}_{k+1}^* = \mathbf{B} \mathbf{V}^n + \theta \mathbf{s}(I_{\text{ion}}(\mathbf{x}_k)) + (1 - \theta) \mathbf{s}^n, \quad (12)$$

followed by a damped update $\mathbf{x}_{k+1} = (1 - \omega) \mathbf{x}_k + \omega \mathbf{x}_{k+1}^*$ with $\omega = \text{nonlinearRelaxation}$. The ionic ODE is re-integrated from t^n to t^{n+1} at every k using $\mathbf{x}_k$ as the (frozen) input voltage.

5.2 Diagonal-Ion linearised Picard

When `nonlinearMethod` is `diagonalIion`, `diagonal` or `localDiagonal`, the implicit operator is augmented by the diagonal of $\partial I_{\text{ion}}/\partial V_m$ at the current iterate, fed back to the implicit side:

$$[A - \theta \text{diag}(\partial I_{\text{ion}}/\partial V_m)_k] \mathbf{x}_{k+1}^* = \mathbf{r}(I_{\text{ion}}(\mathbf{x}_k)) - \theta \text{diag}(\partial I_{\text{ion}}/\partial V_m)_k \mathbf{x}_k, \quad (13)$$

where the diagonal Jacobian entry is approximated by a one-sided finite difference,

$$\left. \frac{\partial I_{\text{ion}}}{\partial V_m} \right|_i \approx \frac{I_{\text{ion}}(V_{m,i} + \varepsilon) - I_{\text{ion}}(V_{m,i})}{\varepsilon}, \quad \varepsilon = \text{diagonalIionEpsilon}. \quad (14)$$

This branch is strictly more accurate than plain Picard on stiff sources and at the same time strictly cheaper than a true Newton solve, because the off-diagonal coupling between cells is left to the implicit diffusion operator only.

5.3 Convergence criteria

The outer loop is allowed up to `nonlinearIterations` sweeps and declared converged when all of the following hold simultaneously:

$$\begin{aligned} \|\mathbf{F}(\mathbf{x}_k)\|_2 / \|\mathbf{x}_k\|_2 &\leq \text{nonlinearTolerance}, \\ \|\mathbf{V}_k - \mathbf{V}_{k-1}\|_2 / \|\mathbf{V}_k\|_2 &\leq \text{nonlinearVmTolerance}, \\ \|I_{\text{ion},k} - I_{\text{ion},k-1}\|_2 / \|I_{\text{ion},k}\|_2 &\leq \text{nonlinearIionTolerance}, \\ (\text{opt.}) \quad \max_i \|\mathbf{s}_k^{(i)} - \mathbf{s}_{k-1}^{(i)}\|_2 / \|\mathbf{s}_k^{(i)}\|_2 &\leq \text{nonlinearStatesTolerance}, \end{aligned}$$

plus $k+1 \geq \text{minNonlinearIterations}$; the last criterion is gated by `nonlinearRequireStatesConvergence`.

5.4 Rollback on non-convergence

If the outer loop exhausts its budget without satisfying the convergence criteria, the user-controlled switch `nonlinearAcceptUnconverged` decides:

- `false` (default): restore $V_m, I_{\text{ion}}, \mathbf{s}$ to the values at t^n and emit a warning;
- `true`: accept the last iterate and continue.

6 Linear solver: Eigen back-end

The wrapper `solveSparseSystem` drives either of two Eigen sparse solvers on the (optionally diagonal-augmented) operator:

linearSolver key	Eigen type	Kind
<code>SparseLU</code>	<code>Eigen::SparseLU</code>	Direct LU factorisation
<code>BiCGSTAB</code>	<code>Eigen::BiCGSTAB + IncompleteLUT</code>	Iterative

Both share the `tolerance` and `maxIterations` dictionary keys (the former is ignored by `SparseLU`). The Picard branch typically uses `SparseLU` so the same factorisation is reused at every outer iteration of the same time step.

7 Ionic ODE integration

For each cell or cell QP, the TNNP ODE (3) is advanced by the OpenFOAM adaptive integrator selected via the `solver` key in `timeIntegrationProperties`. Default `RKF45` [3]. The keys read from that file are:

Key	Meaning	Default
<code>solver</code>	ODE integrator	RKF45
<code>absTol</code>	Absolute tolerance per state	10^{-9}
<code>relTol</code>	Relative tolerance per state	10^{-6}
<code>maxSteps</code>	Cap on sub-steps per outer call	10^4
<code>initialODEStep</code>	Initial sub-step (ms)	10^{-5}

8 TNNP numerical protection

The master switch `TNNPNumericalProtection` (default `false`) enables a guard `protectState(state, t, qp, phase)` invoked at every external entry into the TNNP kernel (`calculateCurrent`, `solveODE_start`, `solveODE_end`, `derivatives`). Knobs:

Key	Type	Default
<code>TNNPNumericalProtection</code>	Switch	<code>false</code>
<code>TNNPCLampGates</code>	Switch	<code>true</code>
<code>TNNPCLampVmForRates</code>	Switch	<code>true</code>
<code>TNNPLogNumericalProtection</code>	Switch	<code>true</code>
<code>TNNPConcentrationFloor</code>	scalar (mM)	10^{-12}
<code>TNNPVMInForRates</code>	scalar (mV)	-200
<code>TNNPVMMaxForRates</code>	scalar (mV)	$+100$
<code>TNNPVoltageZeroEps</code>	scalar (mV)	10^{-8}
<code>TNNPMaxProtectionLogEntries</code>	label	10^6

When enabled, the guards are: (1) reset non-finite V to -86.2 mV; (2) clamp V to $[TNNPVMInForRates, TNNPVMMaxForRates]$; (3) push V off zero by `TNNPVoltageZeroEps` (avoids the i_{CaL} removable singularity); (4) floor K_i, Na_i, Ca_i, Ca_{SR} at `TNNPConcentrationFloor`; (5) clamp gating variables to $[0, 1]$. Corrections are accumulated by `(phase, variable, reason)` and emitted on exit to `TNNP_numericalProtection_summary.dat`.

9 Activation tracking and benchmarks

Activation time per cell. First-crossing of V_{thr} in a step is recorded by linear interpolation:

$$w = \text{clip}\left(\frac{V_{thr} - V^n}{V^{n+1} - V^n}, 0, 1\right), \quad t_{act} = t^n + w \Delta t. \quad (15)$$

The per-cell flag is cleared so re-crossings are ignored.

Early termination on P8. When `stopAfterPointActivation = true`, the cell nearest `stopActivationPoint` (P8 in the Niederer benchmark) is monitored, and the effective end time is set to $t_{act} + \text{stopDelayAfterActivation}$.

Diagonal sampling. `sampleIDW` performs inverse-distance-weighted interpolation at `nDiagonalSamples` points between `diagonalStart` and `diagonalEnd` with $k = \text{sampleNeighbours}$, $p = \text{samplePower}$.

10 Solver flow chart

The pseudo-code below summarises the per-time-step flow and shows the `run_convergence.py constants` / `dict keys` that control each branch.

Algorithm 1 High-level time-step loop of `highOrderElectroActivationFoamImplicit`.

```
1: Read dictionaries.
2: Assemble Eigen SparseMatrixs: M (massMatrix), K (useHighOrderVm), L = K/( $\chi C_m$ ).
3:  $V^0 \leftarrow -84$  mV (initial condition).
4: while  $t^n < \text{END\_TIME}$  do
5:    $\Delta t \leftarrow \text{DT\_VALUES}$  (current sweep value).
6:   Build A = M/ $\Delta t$  -  $\theta$  L, B = M/ $\Delta t$  + (1 -  $\theta$ ) L (implicitScheme  $\rightarrow \theta$ ).
7:   Apply stimulus to  $I_{\text{ext}}$  (stimulusProtocol).
8:    $x_0 \leftarrow V^n$ .
9:   for  $k = 0, 1, \dots, \text{IMPLICIT\_NONLINEAR\_ITERATIONS}-1$  do
10:    Reconstruct  $V_m$  at quadrature points if useHighOrderIion.
11:    Integrate TNNP ODE with  $x_k$  (ODE_REL_TOL, ODE_ABS_TOL, solver=RKF45).
12:    Build RHS:  $r_k = B V^n + \theta s(I_{\text{ion}}(x_k)) + (1 - \theta) s^n$ .
13:    if nonlinearMethod = diagonalIion then
14:       $A_k \leftarrow A - \theta \text{diag}(\partial I_{\text{ion}} / \partial V_m)_k$  (one-sided FD, diagonalIionEpsilon).
15:       $r_k \leftarrow r_k - \theta \text{diag}(\partial I_{\text{ion}} / \partial V_m)_k x_k$ .
16:    else
17:       $A_k \leftarrow A$ .
18:    end if
19:    Solve  $A_k V_* = r_k$  with Eigen (linearSolver, tolerance, maxIterations).
20:     $x_{k+1} \leftarrow (1 - \omega) x_k + \omega V_*$ ,  $\omega = \text{nonlinearRelaxation}$ .
21:    if convergence tests pass (nonlinearTolerance, ...) then
22:      break
23:    end if
24:  end for
25:   $V^{n+1} \leftarrow x_{k+1}$ .
26:  if not converged and not nonlinearAcceptUnconverged then
27:    Warn; roll back:  $V^{n+1} \leftarrow V^n$ ,  $I_{\text{ion}}^{n+1} \leftarrow I_{\text{ion}}^n$ ,  $s^{n+1} \leftarrow s^n$ .
28:  end if
29:  Record activation times (activationThreshold); P8 stop check (stopAfterPointActivation, stopDelayAfterActivation).
30:  Write nonlinear-residual log if writeNonlinearResiduals.
31:   $t^{n+1} \leftarrow t^n + \Delta t$ .
32: end while
33: Post-process: IDW sampling at P8 and diagonal (nDiagonalSamples, sampleNeighbours, samplePower).

---


```

Legend. `Blue` = Python constant set in `run_convergence.py`; `green` = OpenFOAM dictionary key read by the solver.

11 Driver script: `run_convergence.py`

11.1 Geometry and parameter sweep

Python constant	OpenFOAM key	Meaning
DIMENSIONS	—	List of dimensions to sweep (2D, 3D).
DX_VALUES_MM	<code>blockMeshDict</code> cell counts	Uniform cell size in millimetres.
DT_VALUES	<code>controlDict.deltaT</code>	Outer time-step sizes (seconds).
END_TIME	<code>controlDict.endTime</code>	End time (s).
CFL	<code>timeIntegrationProperties.CFL</code>	Informational reference; not enforced (the solver is implicit).
ACTIVE_TIME_SCHEMES	<code>spatialIntegrationProperties</code> { <code>backwardScheme</code> , <code>crankNicolson</code> }.	

Python constant	OpenFOAM key	Meaning
ACTIVE_MASS_MATRICES	spatialIntegrationProperties.massMatrixConsistent	$\{ \text{true}, \text{false} \}$.
ACTIVE_HO_PAIRS_Vm_States	useHighOrder_Vm	/ Tuples (Vm-level, Iion-level); + level $\in \{ \text{N0}, \text{p1}, \text{p2}, \text{p3} \}$.
	useHighOrder_Iion	
	LRECoeffs_*	
ACTIVE_NONLINEAR_METHODS	nonlinearMethod	$\{ \text{picard}, \text{diagonalIion} \}$.
ACTIVE_STATE_ODE_SOLVERS	timeIntegrationProperties.ode	ODE integrator names (RK45, ...).
SIGMA_X, SIGMA_Y, SIGMA_Z	cardiacProperties	Diagonal of D .
CHI	cardiacProperties.chi	Surface-to-volume ratio.
CM	cardiacProperties.Cm	Membrane capacitance.

11.2 Nonlinear and linear solver

Python constant	OpenFOAM key	Meaning
IMPLICIT_NONLINEAR_ITERATIONS	nonlinearIterations	Picard iteration cap per step.
IMPLICIT_MIN_NONLINEAR_ITERATIONS	minNonlinearIterations	Minimum sweeps.
NONLINEAR_TOLERANCE	nonlinearTolerance	Coupled-residual tolerance.
NONLINEAR_VM_TOLERANCE	nonlinearVmTolerance	Relative L^2 change in V_m .
NONLINEAR_IION_TOLERANCE	nonlinearIionTolerance	Same on I_{ion} .
NONLINEAR_STATES_TOLERANCE	nonlinearStatesTolerance	Max over states.
NONLINEAR_REQUIRE_STATES_CONVERGENCE	nonlinearRequireStatesConvergence	Gate state test.
NONLINEAR_RELAXATION	nonlinearRelaxation	Picard mix factor ω .
DIAGONAL_IION_EPSILON	diagonalIionEpsilon	FD step for diagonal $\partial I_{\text{ion}} / \partial V_m$ when $\text{nonlinearMethod} = \text{diagonalIion}$.
IMPLICIT_LINEAR_SOLVER	linearSolver	$\{ \text{SparseLU}, \text{BiCGSTAB} \}$.
IMPLICIT_TOLERANCE	tolerance	BiCGSTAB relative tolerance.
IMPLICIT_MAX_ITERATIONS	maxIterations	BiCGSTAB iteration cap.

11.3 Inner ODE integrator (RK45)

Python constant	OpenFOAM key	Meaning
ODE_ABS_TOL	timeIntegrationProperties.absTol	Absolute tolerance per state.
ODE_REL_TOL	timeIntegrationProperties.relTol	Relative tolerance per state.
ODE_INITIAL_STEP	initialODEStep	Initial sub-step (ms).
ODE_MAX_STEPS	maxSteps	RKF45 sub-step cap.

11.4 Post-processing and run-control

Python constant	OpenFOAM key	Meaning
ACTIVATION_THRESHOLD	activationThreshold	V_{thr} (V).
N_DIAGONAL_SAMPLES	nDiagonalSamples	IDW samples.
SAMPLE_NEIGHBOURS	sampleNeighbours	k -NN for IDW.
SAMPLE_POWER	samplePower	Inverse-distance power.
STOP_AFTER_POINT_ACTIVATION	stopAfterPointActivation	Early-stop on P8.
STOP_DELAY_AFTER_ACTIVATION	stopDelayAfterActivation	Delay (s).

Python constant	OpenFOAM key	Meaning
WRITE_NONLINEAR_RESIDUALS	writeNonlinearResiduals	Residual log.
SAVE_LOGS_ON_SUCCESS	—	Python-only; keep logs after successful run.

12 Niederer benchmark results

This section collects measured P8 activation times, wall-clock cost and peak memory from four independent result trees of this solver:

1. `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderNiedererEtAl2012Implicit`
1/
3. `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderNiedererEtAl2012Implicit`
3/
4. `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderNiedererEtAl2012Implicit`
4/

Rows of the tables below are LRE p-levels for $V_m \times I_{\text{ion}}$ (NO = high-order disabled); columns are mesh sizes Δx in millimetres. Computational cost is the elapsed wall-clock time of the solver process reported by `/usr/bin/time -v`; peak memory is the maximum resident set size of the same process. The nonlinear method, time-integration scheme and mass matrix are noted above each block when relevant. NC = no run data found for that combination.

All runs use the Crank–Nicolson scheme with the consistent high-order mass matrix.

12.1 Results for $\Delta t = 1.0 \cdot 10^{-3}$ s

Table 5: P8 activation [ms] at $\Delta t = 1.0 \cdot 10^{-3}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	192.75	99.19	92.01	91.96	91.51
NO	p1	119.65	92.56	92.16	92.10	91.51
NO	p2	76.52	87.59	91.98	92.11	91.51
NO	p3	76.63	88.19	91.99	92.12	91.51
p1	NO	0.00	109.58	93.32	92.33	91.81
p1	p1	145.47	100.72	94.02	92.70	91.80
p1	p2	82.36	91.27	93.30	93.14	NC
p1	p3	83.96	91.90	93.30	93.15	NC
p2	NO	0.00	127.89	97.27	93.22	92.29
p2	p1	192.75	114.03	96.97	93.64	92.27
p2	p2	87.99	93.26	94.17	93.29	91.98
p2	p3	94.22	94.32	94.11	93.29	NC
p3	NO	0.00	110.29	91.18	90.23	91.51
p3	p1	121.21	97.00	92.11	90.79	91.46
p3	p2	75.08	87.35	91.44	91.29	91.31
p3	p3	71.20	88.09	91.41	91.32	91.31

Table 6: Wall-clock time [s] at $\Delta t = 1.0 \cdot 10^{-3}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	44.7	103.1	378.9	599.7	2316.9
NO	p1	58.8	203.2	770.4	1157.8	4665.2

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	p2	108.3	485.5	2331.7	3257.5	14510.4
NO	p3	143.9	672.2	3117.3	4419.3	16542.8
p1	NO	66.6	118.0	378.8	562.9	2480.0
p1	p1	75.6	202.6	753.3	1083.5	4983.1
p1	p2	127.4	615.6	2149.8	3505.2	NC
p1	p3	166.4	798.5	2801.1	5025.6	NC
p2	NO	75.2	133.2	406.9	640.9	2765.5
p2	p1	94.8	240.0	782.8	1175.1	7091.9
p2	p2	127.8	559.5	2414.4	5070.7	14762.5
p2	p3	192.3	778.3	3161.2	4880.0	NC
p3	NO	54.9	118.4	399.2	657.1	2629.8
p3	p1	60.3	214.9	769.6	1219.4	4863.6
p3	p2	108.0	498.9	2283.6	3453.2	14224.8
p3	p3	142.7	687.7	2987.8	5289.8	18556.7

Table 7: Peak RSS [MiB] at $\Delta t = 1.0 \cdot 10^{-3}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	148	154	178	196	362
NO	p1	150	158	195	226	482
NO	p2	155	176	269	341	946
NO	p3	157	184	306	400	1177
p1	NO	152	166	233	279	687
p1	p1	153	170	248	308	805
p1	p2	157	188	321	423	NC
p1	p3	159	195	358	481	NC
p2	NO	152	169	260	329	852
p2	p1	154	174	278	360	975
p2	p2	158	193	348	473	1429
p2	p3	160	202	390	531	NC
p3	NO	155	192	345	466	1370
p3	p1	157	196	361	485	1465
p3	p2	161	214	437	608	1923
p3	p3	163	223	474	666	2155

12.2 Results for $\Delta t = 1.0 \cdot 10^{-4}$ s

Table 8: P8 activation [ms] at $\Delta t = 1.0 \cdot 10^{-4}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	161.05	69.88	53.92	52.21	50.29
NO	p1	96.18	61.22	52.11	51.29	50.20
NO	p2	37.79	46.32	48.97	49.42	49.93
NO	p3	38.29	46.61	48.91	49.38	49.93
p1	NO	17.70	77.32	57.82	55.10	51.10
p1	p1	101.11	67.56	55.66	53.96	51.01
p1	p2	41.45	50.78	51.32	51.27	NC
p1	p3	43.36	51.34	51.27	51.23	NC
p2	NO	0.00	93.48	64.47	59.77	52.18
p2	p1	162.02	78.77	60.67	57.55	51.98
p2	p2	46.23	53.99	52.87	52.50	51.07
p2	p3	52.60	55.56	53.00	52.53	NC
p3	NO	189.23	76.54	58.33	55.11	50.64
p3	p1	72.53	65.88	55.21	53.43	50.52
p3	p2	37.84	46.86	49.57	49.90	49.94

Vm	Iion	0.5	0.25	0.125	0.1	0.05
p3	p3	39.03	47.38	49.56	49.88	49.94

Table 9: Wall-clock time [s] at $\Delta t = 1.0 \cdot 10^{-4}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	45.5	77.7	245.4	392.4	1518.7
NO	p1	57.8	134.8	479.3	729.1	2856.6
NO	p2	66.0	295.1	1425.8	1960.3	8686.4
NO	p3	88.0	400.2	1864.8	2700.8	10687.5
p1	NO	62.5	101.3	363.8	452.0	2091.9
p1	p1	69.0	160.7	637.0	786.8	3804.2
p1	p2	74.2	381.5	1373.3	2270.6	NC
p1	p3	99.7	507.0	1766.0	3115.6	NC
p2	NO	93.3	136.1	407.1	634.3	3530.4
p2	p1	104.2	202.0	664.6	954.7	4825.6
p2	p2	85.3	375.2	1599.4	2987.3	10193.3
p2	p3	120.5	544.7	2078.0	3139.2	NC
p3	NO	70.7	123.7	460.6	728.8	3061.3
p3	p1	81.1	180.2	612.9	1062.4	4602.5
p3	p2	68.8	321.6	1412.6	2359.3	10293.6
p3	p3	91.2	442.4	2044.3	3226.4	13916.8

Table 10: Peak RSS [MiB] at $\Delta t = 1.0 \cdot 10^{-4}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	149	154	178	196	362
NO	p1	150	158	195	226	481
NO	p2	155	176	269	341	946
NO	p3	158	184	306	399	1177
p1	NO	152	166	234	279	687
p1	p1	153	170	247	308	801
p1	p2	157	187	320	422	NC
p1	p3	160	195	357	479	NC
p2	NO	152	169	258	327	882
p2	p1	154	174	277	357	969
p2	p2	158	192	345	470	1423
p2	p3	160	202	387	528	NC
p3	NO	155	191	344	463	1371
p3	p1	157	196	365	482	1457
p3	p2	161	213	434	605	1912
p3	p3	163	222	472	663	2142

12.3 Results for $\Delta t = 1.0 \cdot 10^{-5}$ s

Table 11: P8 activation [ms] at $\Delta t = 1.0 \cdot 10^{-5}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	157.69	67.69	50.65	48.55	46.10
NO	p1	94.67	59.19	48.68	47.50	46.00
NO	p2	35.26	43.20	45.15	45.42	45.69

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	p3	36.01	43.73	45.12	45.39	45.69
p1	NO	17.33	74.83	54.52	51.58	47.05
p1	p1	99.06	65.26	52.26	50.31	46.93
p1	p2	38.95	47.69	47.65	47.41	NC
p1	p3	41.20	48.40	47.62	47.38	NC
p2	NO	0.00	90.68	61.15	56.52	48.35
p2	p1	159.19	76.35	57.44	54.19	48.08
p2	p2	44.04	51.35	49.47	48.89	NC
p2	p3	50.47	52.95	49.65	48.94	NC
p3	NO	185.33	74.04	54.99	51.79	46.68
p3	p1	71.65	63.54	52.00	50.00	46.51
p3	p2	35.24	43.97	46.03	46.16	45.82
p3	p3	36.80	44.73	46.06	46.16	45.82

Table 12: Wall-clock time [s] at $\Delta t = 1.0 \cdot 10^{-5}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	92.8	166.4	492.7	771.8	3093.3
NO	p1	105.6	272.8	932.9	1422.5	5379.7
NO	p2	121.7	542.6	2622.9	3580.7	14539.8
NO	p3	163.9	737.0	3313.0	5049.9	20077.2
p1	NO	157.9	266.5	1038.3	1293.6	6340.8
p1	p1	148.5	365.7	1512.8	1906.4	9213.2
p1	p2	144.6	1407.1	2758.7	4690.3	NC
p1	p3	197.2	1051.2	3456.0	5984.3	NC
p2	NO	223.7	481.6	1494.2	2134.0	10060.1
p2	p1	258.1	584.2	1996.7	2711.7	12230.0
p2	p2	174.0	836.4	3501.0	5891.3	NC
p2	p3	259.2	1157.2	4163.7	6546.5	NC
p3	NO	230.7	516.7	1887.1	3052.6	11896.2
p3	p1	177.4	607.5	2081.4	3567.7	15532.4
p3	p2	151.6	771.6	3408.6	5846.8	24506.4
p3	p3	191.8	1043.4	4651.7	7839.9	32224.2

Table 13: Peak RSS [MiB] at $\Delta t = 1.0 \cdot 10^{-5}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	149	155	178	196	362
NO	p1	150	158	195	226	481
NO	p2	155	176	269	342	945
NO	p3	157	185	306	399	1177
p1	NO	151	167	233	279	673
p1	p1	153	170	246	311	785
p1	p2	157	188	318	420	NC
p1	p3	160	195	356	477	NC
p2	NO	153	170	259	323	853
p2	p1	154	174	276	352	949
p2	p2	158	193	345	466	NC
p2	p3	161	202	387	524	NC
p3	NO	155	191	342	461	1371
p3	p1	157	196	362	482	1482
p3	p2	162	212	433	603	1893
p3	p3	164	222	470	660	2123

12.4 Results for $\Delta t = 1.0 \cdot 10^{-6}$ s

Table 14: P8 activation [ms] at $\Delta t = 1.0 \cdot 10^{-6}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	157.37	67.46	50.32	48.18	45.69
NO	p1	94.51	58.99	48.34	47.13	45.59
NO	p2	35.04	42.92	44.79	45.04	NC
NO	p3	35.82	43.47	44.76	45.01	NC
p1	NO	17.30	74.57	54.18	51.21	NC
p1	p1	98.86	65.03	51.92	49.94	NC
p1	p2	38.74	47.39	47.29	NC	NC
p1	p3	41.03	48.13	47.27	NC	NC
p2	NO	0.00	90.40	60.81	56.17	NC
p2	p1	158.92	76.11	57.10	53.84	NC
p2	p2	43.86	51.09	49.14	NC	NC
p2	p3	50.29	52.70	49.33	NC	NC
p3	NO	184.97	73.78	54.64	51.44	46.28
p3	p1	71.56	63.31	51.67	49.65	NC
p3	p2	35.01	43.70	45.69	45.80	NC
p3	p3	36.61	44.48	45.73	NC	NC

Table 15: Wall-clock time [s] at $\Delta t = 1.0 \cdot 10^{-6}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	556.1	982.7	2925.8	4661.3	17045.8
NO	p1	619.0	1548.7	5296.0	8197.9	30355.5
NO	p2	650.7	2967.5	12675.5	20197.4	NC
NO	p3	860.9	3979.0	17276.9	30442.0	NC
p1	NO	1115.5	1795.7	8055.0	8704.2	NC
p1	p1	1020.6	2343.0	8784.2	12203.9	NC
p1	p2	794.2	5210.0	16778.1	NC	NC
p1	p3	1081.3	7356.0	21548.8	NC	NC
p2	NO	1609.5	4026.0	13191.0	19449.5	NC
p2	p1	1826.7	4305.0	16538.0	23408.4	NC
p2	p2	1023.1	5360.0	22984.4	NC	NC
p2	p3	1489.8	7407.0	30112.7	NC	NC
p3	NO	1795.5	4411.0	15259.0	23952.2	100244.6
p3	p1	1351.5	4784.0	17203.0	25931.0	NC
p3	p2	900.0	5081.0	24638.8	40049.3	NC
p3	p3	1136.7	6346.0	28125.6	NC	NC

Table 16: Peak RSS [MiB] at $\Delta t = 1.0 \cdot 10^{-6}$ s.

Vm	Iion	0.5	0.25	0.125	0.1	0.05
NO	NO	150	154	178	195	360
NO	p1	150	158	195	225	480
NO	p2	155	176	269	341	NC
NO	p3	157	184	306	399	NC
p1	NO	152	166	231	280	NC
p1	p1	153	170	247	310	NC
p1	p2	157	188	318	NC	NC
p1	p3	160	195	355	NC	NC
p2	NO	153	170	261	324	NC
p2	p1	154	175	277	355	NC

Vm	Iion	0.5	0.25	0.125	0.1	0.05
p2	p2	158	192	344	NC	NC
p2	p3	160	202	386	NC	NC
p3	NO	155	192	342	461	1371
p3	p1	157	197	362	482	NC
p3	p2	161	213	432	602	NC
p3	p3	164	222	469	NC	NC

A Dimension table

Field	OpenFOAM dimensionSet	SI
Vm	[1, 2, -3, 0, 0, -1, 0]	volt
Iion	dimVoltage/dimTime	$V s^{-1}$
externalStimulusCurrent	dimCurrent/dimVolume	$A m^{-3}$
conductivity	dimCurrent^2 dimTime^3 / (dimMass dimVolume)	$S m^{-1}$
chi	dimless/dimLength	m^{-1}
Cm	dimCurrent dimTime / (dimVoltage dimArea)	$F m^{-2}$

References

- [1] K. H. W. J. ten Tusscher, D. Noble, P. J. Noble, A. V. Panfilov, *A model for human ventricular tissue*, AJP – Heart and Circulatory Physiology **286** (2004) H1573–H1589.
- [2] S. A. Niederer et al., *Verification of cardiac tissue electrophysiology simulators using an N-version benchmark*, Prog. Biophys. Mol. Biol. **104** (2012) 22–48.
- [3] E. Hairer, G. Wanner, *Solving Ordinary Differential Equations II*, Springer, 1996.